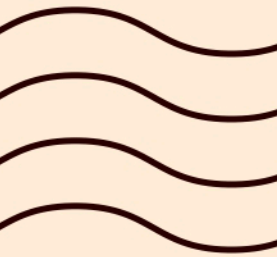




PRESENTS

FINTECHSTICO

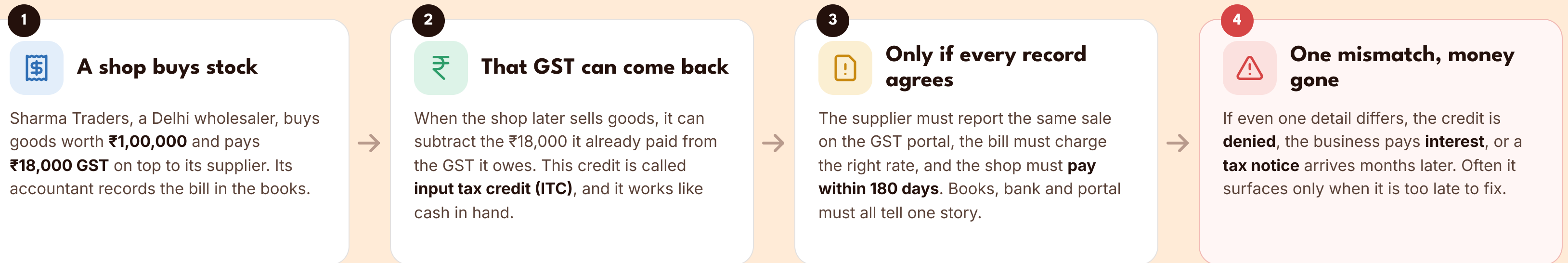
THE FINTECH HACKATHON





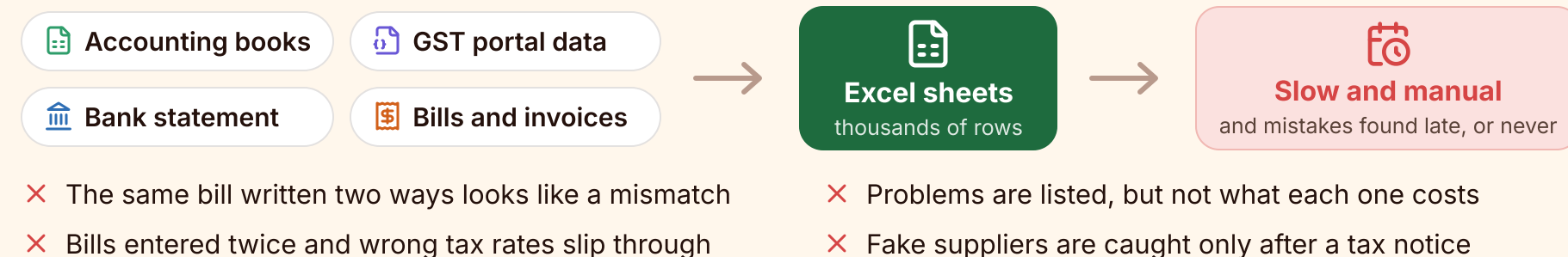
Problem Statement

Businesses lose real money every month because **their records don't match the government's**



Today, an accountant checks this by hand

Every month, four sets of records are downloaded and compared line by line to find what does not match.



1.67 crore

businesses are registered under GST and face this every filing cycle
GSTN, 30 Jun 2026

₹35,132 cr

of fake tax credit caught in just 7 months, much of it through fake supplier bills
CBIC, Apr to Oct 2024

Oct 2025

new rules lock returns to what was already reported, so errors can't be fixed at the last minute
GSTN advisories

Who feels it most: small and mid-sized businesses, and the CA firms that file GST for dozens of them every month.



Team name

Team members



LedgerLens

Your AI assistant for GST checks

M1

Member name

College, year
Role

M2

Member name

College, year
Role

M3

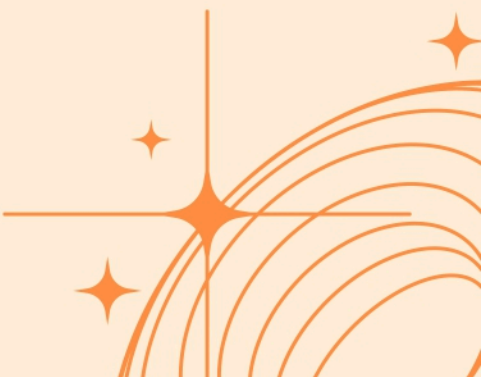
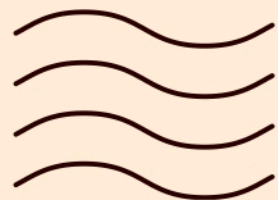
Member name

College, year
Role

M4

Member name

College, year
Role





Solution

LedgerLens checks every bill, payment and tax record, **shows the money at stake**, and tells you what to do

Money at risk

Money to claim



1. Upload

Drop in the accounting books, bank statement and GST portal downloads. LedgerLens reads them as they are, so nobody retypes a single entry.



2. Check

Every bill is matched to its payment, its accounting entry and the supplier's GST filing, even when written differently. The tax rate on each bill is checked too.



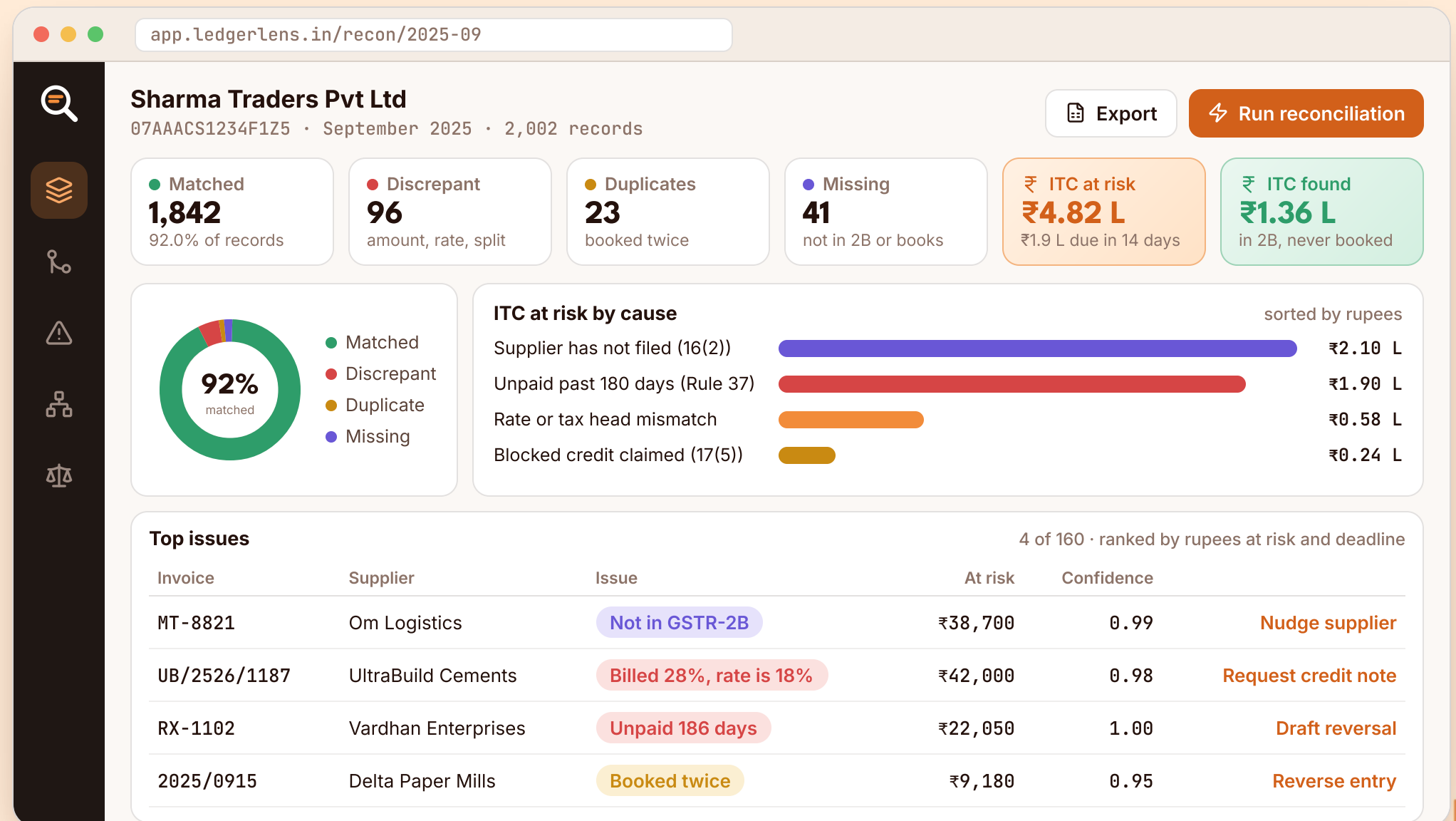
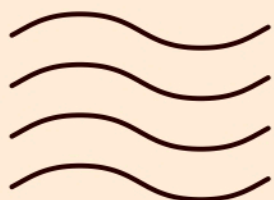
3. Explain

Each problem shows the money involved: credit at risk if left unfixed, or credit owed but never claimed. A plain note says why, with both records as proof.



4. Fix

It drafts the next step, like an email asking the supplier to correct their filing, or the entry that fixes the books. You review and approve in one click.



Covers every requirement in the problem statement

Matches bills and payments

Finds duplicates and mismatches

Checks the tax on every bill

Finds missing records

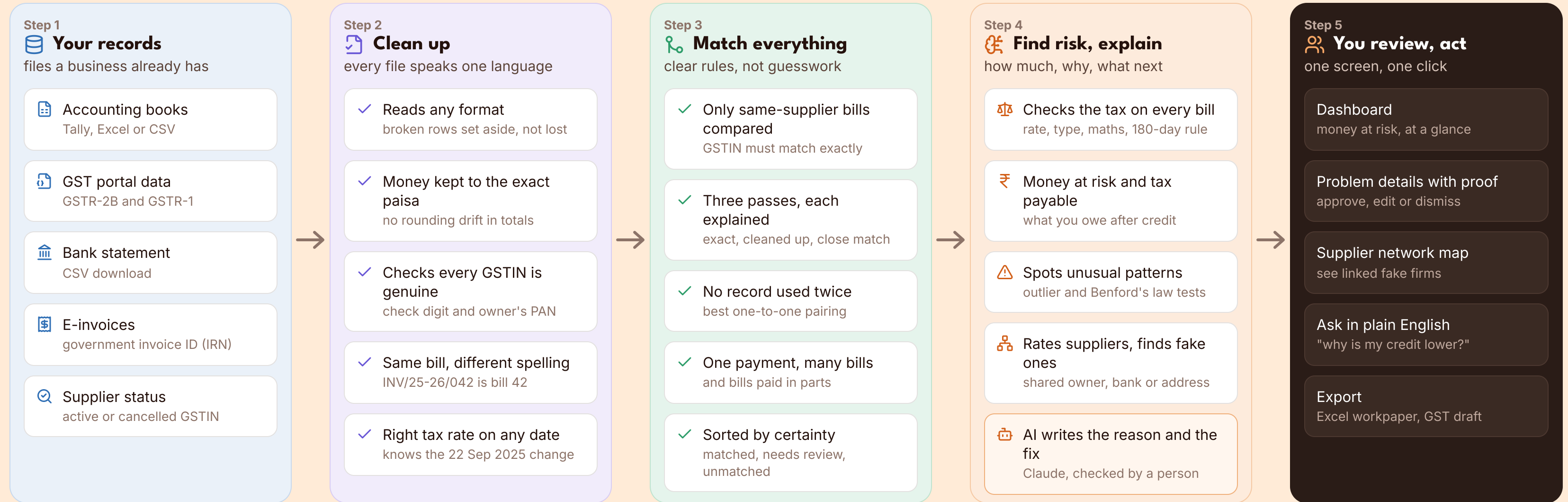
Flags suspicious patterns

Works out tax payable

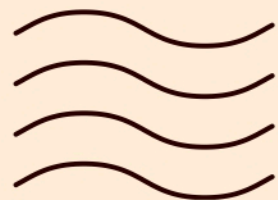
Visual dashboard



Architecture



 **Runs on one laptop** a local database (SQLite) keeps everything together, so the demo works even without internet



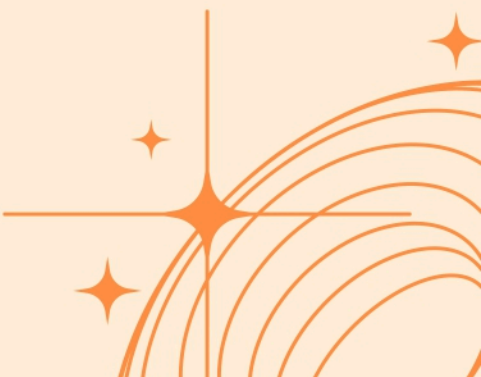
Maths by code, words by AI

Every rupee figure comes from tested rules. The AI only explains and drafts. It can never change a number.



Proven on test data

We plant known mistakes in realistic books and count how many we catch, and how often we raise a false alarm.





Technology Used

Website and dashboard

what users see



Next.js



TypeScript



Tailwind CSS

shadcn/ui clean components

Recharts charts

Cytoscape.js supplier map

Server and data

where records live



Python



FastAPI connects screen and engine



Pydantic checks every file

SQLite local database



Polars fast tables

Matching engine

finds and pairs records

RapidFuzz near-identical bill numbers

SciPy best one-to-one pairing



scikit-learn unusual pattern finder

NetworkX links between suppliers

AI assistant

explains and drafts



Claude API reasons and drafts

Sees only the facts of one problem

Saved answers works offline

Human approves every action

Reliability

so the demo never breaks



Docker runs anywhere

pytest tests for every rule

Playwright clicks through the demo

Offline demo mode

Test data that knows the answers

A full year (FY 2025-26) of books for a fictional trading company. Mistakes are planted on purpose and recorded, so we can count exactly what we catch and what we miss.

5,197

bills and invoices

4,690

bank entries

9,948

accounting entries

781

planted mistakes

30

kinds of mistake

1,368

honest traps

Honest traps look like errors but are fine, such as bills paid in parts. A good system must leave them alone.

Examples of planted mistakes

Bill entered twice

Old tax rate after 22 Sep

Wrong tax type

Tax maths error

Bill number typo

Missing entry

Payment with no bill

Money sent in a circle

Bills split to dodge limits

Invalid GSTIN

Credit over-claimed

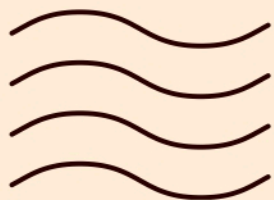
We add supplier filings (GSTR-2B) on top, so missing supplier reports and unclaimed credit can be tested too. Tax rates follow the official CBIC schedule.

Catch rate

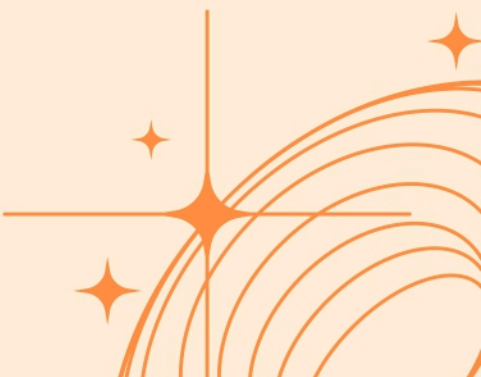
share of real mistakes found

False alarm rate

honest records wrongly flagged



Why these tools: free, proven and widely used. Numbers are handled in Python, where every rule can be tested. Screens are built in TypeScript, where the experience is polished.





Feature / USP



Shows money, not rows

Every problem shows the rupees at risk and the rupees you can still claim, sorted so the nearest deadline comes first. You fix what costs most, first.



Every flag explains itself

Each flag gives a plain-language reason, the GST rule behind it, both records side by side and how sure we are. Anyone can check it, so there is no black box.



Fixes, not just finds

It drafts the supplier email, credit note request or accounting correction for each problem. Nothing is sent until you approve, so you stay in control.



Spots fake suppliers

It links suppliers that share an owner (PAN), bank account or address, revealing money moving in circles before the tax department finds it.

app.ledgerlens.in/issues/UB-2526-1187

Tax rate mismatch

₹42,000 at risk

98% sure

UltraBuild Cements · UB/2526/1187 · 24 Sep 2025

Field	Your books	GSTR-2B	Expected
HSN	2523	2523	2523
Taxable value	4,20,000	4,20,000	4,20,000
Rate	28%	28%	18%
IGST	1,17,600	1,17,600	75,600

Rule: cement moved to 18% under GST 2.0, effective 22 Sep 2025

🚩 Why this was flagged

Cement (HSN 2523) dropped from 28% to 18% on 22 Sep 2025. This invoice is dated 24 Sep but was billed at the old rate. The excess ₹42,000 should be corrected by a credit note before you claim it.

What to do

Ask UltraBuild for a ₹42,000 credit note. Until it arrives, claim credit only on the correct ₹75,600.

✉️ Drafted for you: credit note request

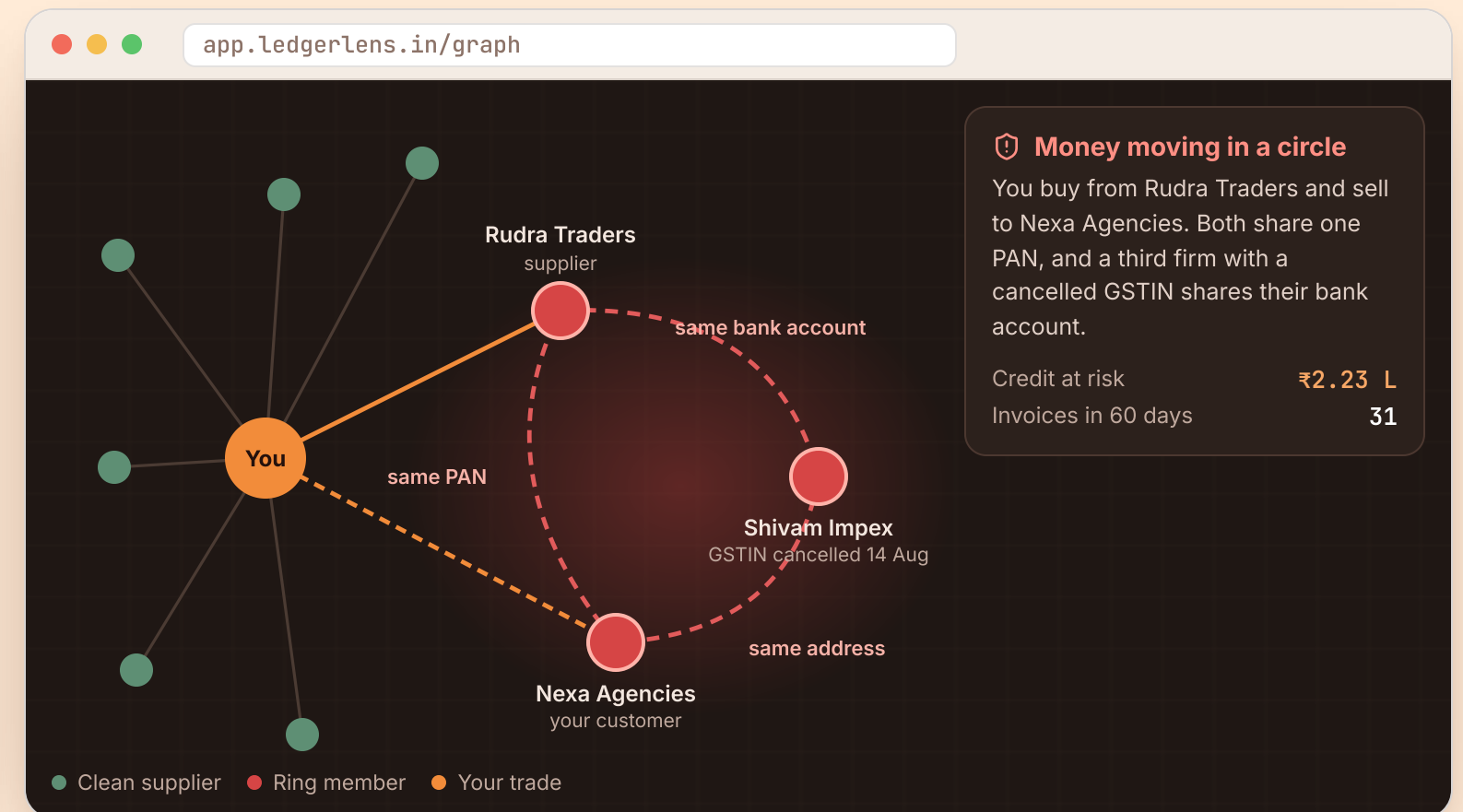
to accounts@ultrabuild.in

Invoice UB/2526/1187 dated 24 Sep 2025 applies 28% IGST to HSN 2523. The rate from 22 Sep 2025 is 18%. Please issue a credit note for ₹42,000 and refile in GSTR-1.

✓ Approve and send

Edit draft

Dismiss with note



Also: it knows the **GST rate change of 22 Sep 2025**, links **one payment to many bills**, **rates every supplier**, and its **accuracy is measured on test data**, not guessed.